

Simarpreet Singh

Student Developer

Portfolio: <https://www.simarpreetsingh.org>

GitHub: <https://github.com/Sim20004>

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Education	Personal Profile
Queen Mary's Grammar School, Walsall Year 9 Student	Student developer with experience in Python, Rust, web development, Linux administration, and self-hosting. Creator of several software projects including a privacy-focused chat platform, a programming language, and an AI-assisted music learning system. Passionate about open source software and learning new technologies.

Technical Skills	
Programming	Technologies
• Python • Rust • HTML • CSS • Java	• Linux • Git • GitHub • Docker • Nginx • Cloudflare

In practice: Python builds the Teapot compiler; Linux, PHP, and MySQL run the self-hosted infrastructure behind Tenberal News; HTML/CSS/JS shape the portfolio site; Git, GitHub, Docker, Nginx, and Cloudflare handle version control and deployment across all of the projects below.

Qualifications & Achievements		
Computer Science	Music	Mandarin
OCR Entry Level Certificate in Computer Science (ELCS) 30/07/2026	ABRSM Grades 1-4 Piano (Practical) with Distinction, Merit, Distinction, and Merit respectively 2023-2026 Trinity College Grade 4 Clarinet (Performance) with Merit 12/02/2026	National Achievement Hanyu Qiao Zhongwen Shuo Bisai (China Bridge National Mandarin Speaking Competition) organised by Hanyu Qiao Organisation (China Bridge) First place in Primary Category nationwide. 26/10/2025

Projects – repositories are linked in portfolio above.

Tenberal News *(Live)*

- Designed a proprietary, professional CRUD news platform and implemented the backend.
- Implemented WebAuthn passkey authentication with cross-browser compatibility fixes
- Built a World Cup statistics dashboard integrating a third-party sports API
- Worked with servers, PHP, Nginx, and Linux to host the site.

TeapotOS *(Ongoing)*

- Wrote an operating system from scratch in Assembly, aiming to support Windows, Linux, and MacOS binaries
- Implemented a small, working, 32-bit kernel and a fully functional bootloader

Teapot Programming Language *(Ongoing)*

- Designed and implemented a custom programming language with lexical analyser, parser, and abstract syntax tree
- Built a 9000+ test (100% code coverage) integration/unit suite covering the compiler pipeline, CLI and individual aspects

Personal Portfolio Site *(Ongoing)*

- Designed and developed a personal website
- Implemented responsive design and SEO optimisation

Technical Experience

- Self-hosted Linux servers
- Docker container deployment and management
- Nginx web server configuration
- Cloudflare DNS and domain management
- Git and GitHub workflows

Leadership and Public Speaking

- Invited to deliver a speech to officials and representatives involved in UK-China relations
- Presented speech in both Mandarin and English

Languages

Punjabi	English	Mandarin
Native speaker	Native speaker	Intermediate proficiency Participant in the Mandarin Excellence Programme (MEP)

Interests

- Software Development
- Open Source Software
- Linux & Self-Hosting
- Programming Language/Compiler Design
- Piano, Clarinet, and Saxophone
- Cybersecurity

